Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 5: Design and Implementation of 4-bit Parallel Binary Adder

Objective:

- To investigate how the Half adder and Full adder circuits work
- To gain experience working with practical circuits.
- To investigate how the IC 7483(4-bit parallel adder) works

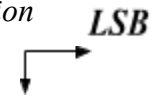
Required Components:

1. IC 7408
2. IC 7432
3. IC 7486
4. IC 7483

Theory:

The addition of two binary numbers is performed in exactly the same manner as the addition of decimal numbers.

Let us first review the decimal addition



3 7 6
4 6 1
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8 3 7

The least significant digit position is operated on first, producing a sum of 7. The digits in the second position are then added to produce a sum of 13, which produces a ***carry*** *of 1 into the third position. This produces a sum of 8 in the third position.*

The same general steps are followed in binary addition. However, only four cases can occur in adding the two binary digits (bits) in any position. They are

0+0=0
1+0=1
1+1=10=0+carry of 1 into the next position
1+1+1=11=1+carry of 1 into the next position

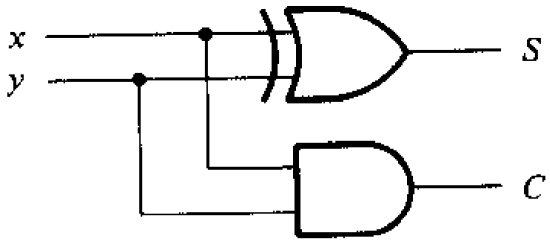
Here are several examples of the addition of two binary numbers:

1001	*1101*
1111	*0110*
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11000	*10011*

Half Adder Circuit:

A half adder is a combinational circuit that forms the arithmetic sum of two input bits. It consists of two inputs and two outputs. Two of the input variables, denoted by x and y represent the two significant bits to be added. The two outputs are designed by the symbols S and C. The binary S gives the value of the least significant bit of the sum. The binary variable C gives the output carry. The truth table of the full adder is as follows:

X	Y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



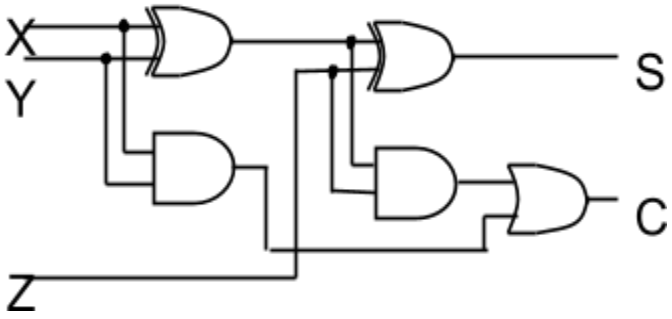
(e) $S = x \oplus y$
$C = xy$

Figure: Half Adder circuit

Full Adder Circuit:

A full adder is a combinational circuit that forms the arithmetic sum of three input bits. It consists of three inputs and two outputs. Two of the input variables, denoted by x and y represent the two significant bits to be added. The third input z represents the carry from the previous lower significant position. The two outputs are designed by the symbols S and C. The binary S gives the value of the least significant bit of the sum. The binary variable C gives the output carry. The truth table of the full adder is as follows:

x	*y*	*z*	*C*	*S*
0	*0*	*0*	*0*	*0*
0	*0*	*1*	*0*	*1*
0	*1*	*0*	*0*	*1*
0	*1*	*1*	*1*	*0*
1	*0*	*0*	*0*	*1*
1	*0*	*1*	*1*	*0*
1	*1*	*0*	*1*	*0*
1	*1*	*1*	*1*	*1*



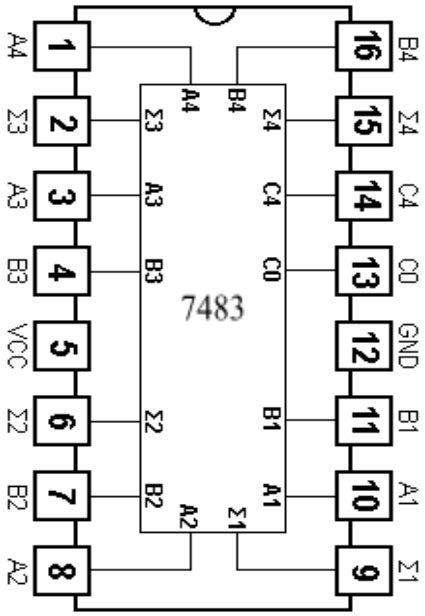
$C = XY + (X \oplus Y)Z$

$S = (X \oplus Y) \oplus Z$

. ***Figure: Full Adder Circuit***

A four-bit parallel adder Circuit (IC - 7483):

A binary parallel adder is a digital function that produces the arithmetic sum of two binary numbers in parallel. It consists of full adders connected in cascade, with the output carry from one full adder connected to the input of the next full adder.



A four bit parallel adder cum subtractor:

IC: 7486(XOR) 7483(4bit parallel adder):

To implement addition and subtraction together:

1. B1 xor C_0 , B2 xor C_0, B3 xor C_0 and B4 xor C_0
2. Connect the output from step 1 to the input of the 7483 IC's B inputs.
3. Keep C_0 common for all steps
4. give C_0= 0 to perform addition, C_0=1 to perform subtraction

We use XOR gate as it produces the invert output of one operand when the other operand is equal to 1.

A	B	Output
1	0	1 (invert of B)
1	1	0 (invert of B)
0	1	1
0	0	0

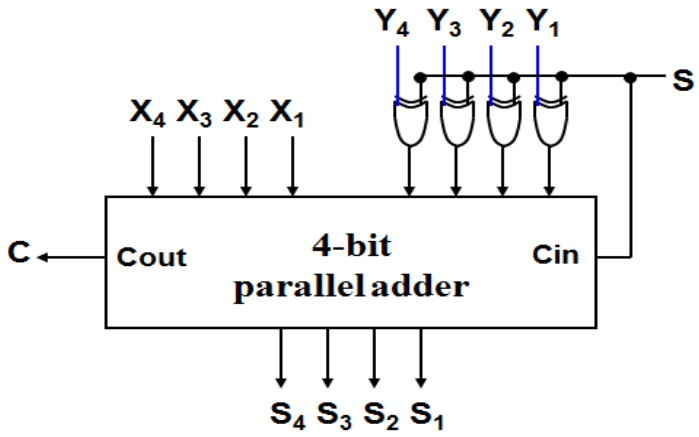


Figure: 4-bit Parallel Adder cum Subtractor

Result:

Fill in the following tables:

(i) Truth table for 4-bit Parallel Adder:

	A	B	Cin	C4	S4	S3	S2	S1
a.	0000	1000	0					
b.	1100	1101	0					
c.	1110	0011	0					
d.	1111	1111	0					

(ii) Truth table for 4-bit Parallel Adder cum Subtractor:

	A	B	Cin	C4	S4	S4	S2	S1
a.	1100	1000	1					
b.	1100	1101	0					
c.	1110	0011	1					
d.	1111	1111	0					